An Ultrasound ASIC With Universal Energy Recycling for >7-m All-Weather Metamorphic Robotic Vision

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***Abstract*—An ultrasound ASIC that enables compact, 3-D nonvisual robotic navigation for >7-m is presented. The proposed ultrasound ASIC mitigates the limitations of camera vision navigation systems in all-weather, low-power, low-cost aspects. The ASIC integrates TX and RX paths for 64 channels, enabling four-directional navigation using 4 × 4 ultrasound arrays. The proposed universal energy recycling transmitter (UERTX) driver circuit, which can drive both single-ended and differential transducers, reduces energy consumption by 44% compared with non-overlap switching-assisted conventional class-D ultrasonic drivers, addressing large TX power consumption. In addition, the on-chip programmable power management unit (PMU) eliminates off-chip supplies and reduces off-chip passive components for compact system miniaturization. Furthermore, PMU enables the tuning of TX driving amplitude for the range of 6–14 V_{pp} allowing wide detection range adaptability. Fabricated in 0.18 μm 1P6M standard CMOS, the ASIC system measures 25 mm^2 and has >7-m obstacle detection capability while consuming 4.3 mW/channel. The proposed ultrasound ASIC system is demonstrated with a four fps, 3-D navigating metamorphic robot, which consumes 0.28 W, occupies 125 cm^3, and weighs ≤100 g.**

***Index Terms*—All-weather, energy recycling, low-power, machine vision, real-time, small form factor, ultrasound.**

I. Introduction

METAMORPHIC robot is a new type of small form factor robot that can change its shape to execute particular tasks, such as pipeline inspection, cave rescue, geological surveys, and building inspections. As its moving speed is slow, a high frame rate is unnecessary, but as it operates in a challenging environment, it must be robust against low-light conditions. For example, a commercial metamorphic robot (Daran Robotics, MW-74Pi) weighs 2 kg with a maximum load capacity of 2 kg, and the form factor is 37 × 25 × 8 cm^3. Therefore, the 3-D vision system mounting on the metamorphic robot should be low power (<1 W), small form factor (∼cm^3), and lightweight (∼0.1 kg). Machine vision is commonly realized using optical means [1], IR, radar, or LIDAR. The visual and IR cameras have high image resolution but require high compute capability and high power consumption to process massive raw image data (>1 GB/s

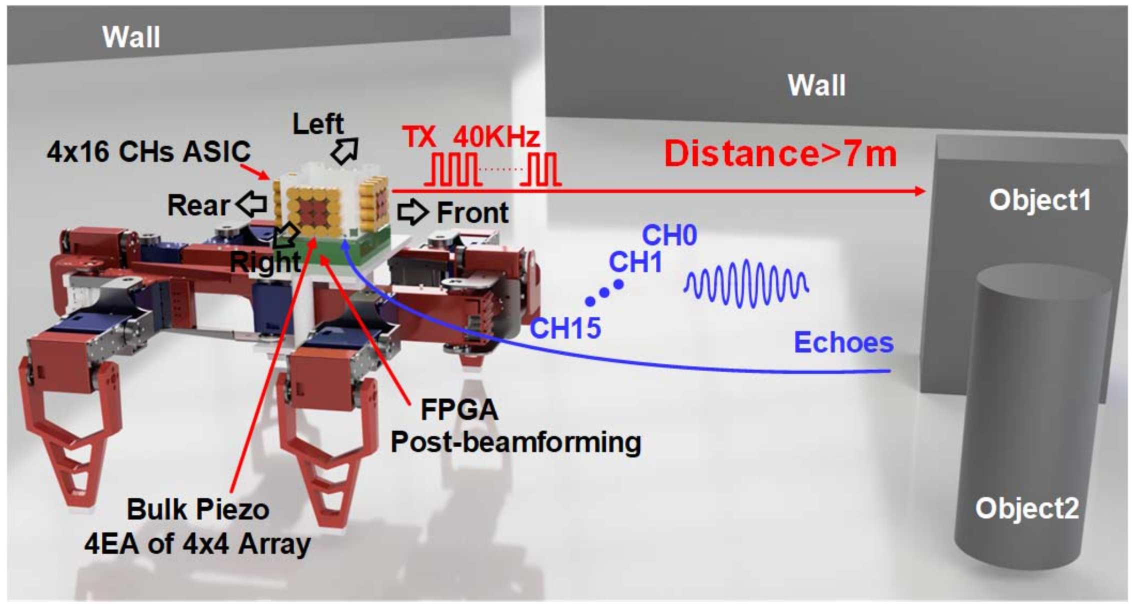

Fig. 1. Ultrasound ASIC for metamorphic robotic vision.